

List of Java Programming Lab Programs based on Basic Java Concepts:

1. Write a Java program to display "Hello, Java!" and demonstrate the basic structure of a Java program.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
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C:\Users\ITLAB2\Desktop>javac HelloJava.java

C:\Users\ITLAB2\Desktop>java HelloJava
Hello Java!

C:\Users\ITLAB2\Desktop>
```

2. Write a Java program to declare variables of different primitive data types (byte, short, int, long, float, double, char, and boolean) and display their values.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2\Desktop>javac DataTypes.java

C:\Users\ITLAB2\Desktop>java DataTypes
Byte: 10
Short: 100
Int: 1000
Long: 100000
Float: 12.5
Double: 123.456
Char: A
Boolean: true

C:\Users\ITLAB2\Desktop>_
```

3. Write a Java program to perform arithmetic operations (addition, subtraction, multiplication, division, and modulus) on two integer variables.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2\Desktop>javac ArithmeticOperations.java

C:\Users\ITLAB2\Desktop>java AritematicOperations
Addition = 25
Subtraction = 15
Multiplication = 100
Division = 4
Modulus = 0

C:\Users\ITLAB2\Desktop>
```

4. Write a Java program to perform arithmetic operations on floating-point numbers and display the results.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
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C:\Users\ITLAB2\Desktop>javac Float.java

C:\Users\ITLAB2\Desktop>java Float
Addition = 13.0
Subtraction = 8.0
Multiplication = 26.25
Division = 4.2

C:\Users\ITLAB2\Desktop>_
```

5. Write a Java program to demonstrate character data type by displaying a character and its corresponding ASCII/Unicode value.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2\Desktop>javac Character.java

C:\Users\ITLAB2\Desktop>java Character
Character = A
ASCII/Unicode Value = 65

C:\Users\ITLAB2\Desktop>
```

6. Write a Java program to demonstrate the use of Boolean variables and Boolean expressions.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
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C:\Users\ITLAB2\Desktop>javac Boolean.java

C:\Users\ITLAB2\Desktop>java Boolean
isJavaEasy = true
10 > 5 = true

C:\Users\ITLAB2\Desktop>_
```

7. Write a Java program to declare and initialize different types of variables and display their values.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
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C:\Users\ITLAB2\Desktop>javac Variable.java

C:\Users\ITLAB2\Desktop>java Variable
Name = Rahul
Age = 20
Salary = 25000.5
Grade = A
Passed = true

C:\Users\ITLAB2\Desktop>
```

8. Write a Java program to swap the values of two variables using a temporary variable.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2\Desktop>javac SwapNumbers.java

C:\Users\ITLAB2\Desktop>java SwapNumbers
Before Swap:
a = 10
b = 20
After Swap:
a = 20
b = 10

C:\Users\ITLAB2\Desktop>
```

9. Write a Java program to demonstrate automatic (widening) type conversion between primitive data types.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
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C:\Users\ITLAB2\Desktop>javac Implicit.java

C:\Users\ITLAB2\Desktop>java Implicit
Integer = 100
Double = 100.0

C:\Users\ITLAB2\Desktop>_
```

10. Write a Java program to demonstrate explicit (narrowing) type casting from one primitive data type to another.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2\Desktop>javac Explicit.java

C:\Users\ITLAB2\Desktop>java Explicit
Double = 123.45
Integer = 123

C:\Users\ITLAB2\Desktop>
```

11. Write a Java program to convert a character to its ASCII/Unicode value and vice versa using type casting.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\ITLAB2\Desktop>javac CharConversion.java

C:\Users\ITLAB2\Desktop>java CharConversion
Character = B
ASCII = 66
ASCII = 67
Character = C

C:\Users\ITLAB2\Desktop>
```

12. Write a Java program to check whether a given number is even or odd using the `if-else` control statement.

```
PS C:\Users\gowth\OneDrive\Desktop\java> javac ifelse.java
PS C:\Users\gowth\OneDrive\Desktop\java> java ifelse
32443423 is odd number
PS C:\Users\gowth\OneDrive\Desktop\java> |
```

13. Write a Java program to find the largest of two numbers using the `if-else` statement.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\gowth\OneDrive\Desktop\java> javac largest.java
PS C:\Users\gowth\OneDrive\Desktop\java> java largest
540 is the larger number than 320
PS C:\Users\gowth\OneDrive\Desktop\java> |
```

14. Write a Java program to demonstrate the use of Java reserved keywords by creating valid identifiers and explaining why keywords cannot be used as variable names.

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
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C:\Users\ITLAB2\Desktop> javac Keywords.java

C:\Users\ITLAB2\Desktop> java Keywords
Number = 10
Name = Java
Status = true

C:\Users\ITLAB2\Desktop>
```

15. Write a Java program to demonstrate the use of the for loop by displaying numbers from 1 to 10.

Output:

```
PS C:\Users\gowth\OneDrive\Desktop\java> javac looping.java
PS C:\Users\gowth\OneDrive\Desktop\java> java looping
1
2
3
4
5
6
7
8
9
10
PS C:\Users\gowth\OneDrive\Desktop\java> |
```